

On The History of the Nocturnal

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Summary

The purpose of the nocturnal is finding the time by night from observation of the fixed stars. Contrary to the common planispheric astrolabe, it is a universal instrument, i.e. it can be used in any latitude. Nocturnals were very popular time-telling devices until the 18th century and were particularly used by sailors. The paper deals with the history of this instrument and its literary tradition (manuscripts and textbooks), which has not been given much attention.

Introduction

Any method of time-reckoning is based on the fact that the positions of the sun and the fixed stars are changing continually in respect to the plane of the horizon which is a result of the earth's rotation. If one watches the northern sky in a clear night it will appear that all stars rotate anticlockwise about the North Pole of the heavens near the Pole Star. The starry sphere completes one revolution in 23 hours and 56 minutes and so performs like a gigantic clock. To read off time is not easy however, because the stars move counterclockwise and sidereal time is different from local mean time. Apparently the sun moves about 1° daily in respect to the zodiacal signs and therefore the mean solar day is not equal to the mean sidereal day, but about 4 minutes longer.

The Nocturnal

The Nocturnal, sometimes also called Night or Star Clock (*Nachtuhr*, *Sternenuhr*) uses the Pole Star and conspicuous stars nearby, especially the 'Guards' of the Great Bear (α und β Ursae maioris, *Dubhe* and *Merak*) for time-telling. The extension of the line of their connection directly points to the Pole Star. Sometimes also the brightest star of the Lesser Bear (β Ursae minoris; *Kochab*) is used. It consists of three concentric plates, which are fixed together, but may be rotated smoothly. The outermost plate is divided counterclockwise into twelve equal parts representing the months, with subdivisions into five-day periods. If the instrument is designed to be used with several stars, the calendar disk is mounted moveable on a ground-plate, because the date on which the respectable star to be used for time-telling has the same rectascension as the sun must lie on the axis of symmetry of the instrument, i.e. the Meridian line. The ground-plate or the calendar disk itself carries a handle so that the instrument can be held easily.

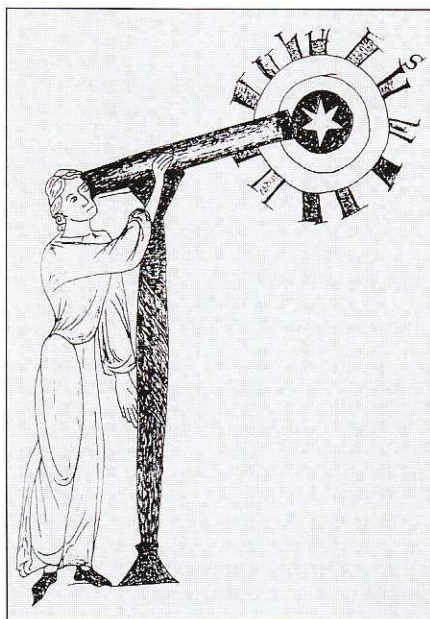


Fig.1 Venedig, Biblioteca Nazionale Marciana, Ms. Lat. VIII. 22, fol. 1^r.

The circumference of the middle plate is divided into twenty-four equal parts, each part corresponding to one hour. To facilitate reading off the hour by night 24 teeth are cut out (the 12-hour marks being longer or differently formed), so that the hour can be made out by touching. The inner plate carries an alidade, which protrudes over the outer rim of the calendar plate. The rivet in the centre fixing together all parts is drilled and serves as a sighting hole for the Pole Star.

To use the nocturnal the 12-hour mark of the middle disk is set to coincide with the respective date, then the instrument is held at arms length and the Pole Star is observed through the hole at the centre. Now the alidade is rotated until its bevelled edge coincides with the Guards and the number of hours can be read off directly or (in darkness) by counting the number of teeth between the 12-hour mark and the edge of the alidade. It should be taken into consideration that the 12-hour mark which is set on the respective date shows midnight. If the hours are counted clockwise one gets the hours after midnight and if counted anticlockwise the hours before midnight are obtained. Because the Pole Star is at present at a distance of about 1° from the Celestial Pole, a certain error is unavoidable, but with a well-executed nocturnal time can be measured within 5-10 minutes.

Most certainly the nocturnal is of European origin. In a series of papers (1947/52, 1959) Henri Michel postulated

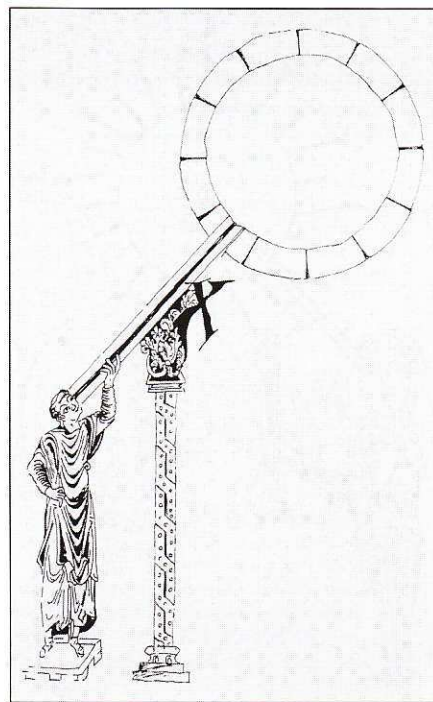


Fig.2 St. Gallen, Stiftsbibliothek, Cod. Sang. 18, p. 45.

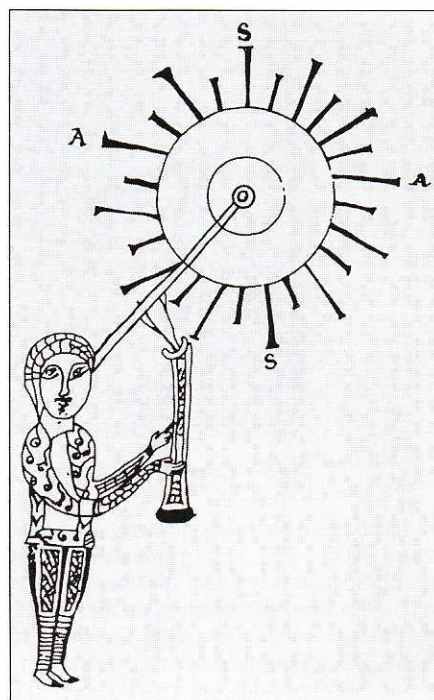


Fig.3 Rome, Bibliotheca Apostolica Vaticana, Vat. Lat. 644, fol. 76r.

its invention back to the 10th century B.C. in China.¹ Michel assumed that the outlines of flat jade disks with three deep notches cut at 120° from each other and with three series of serrations between these teeth fitted closely to the positions

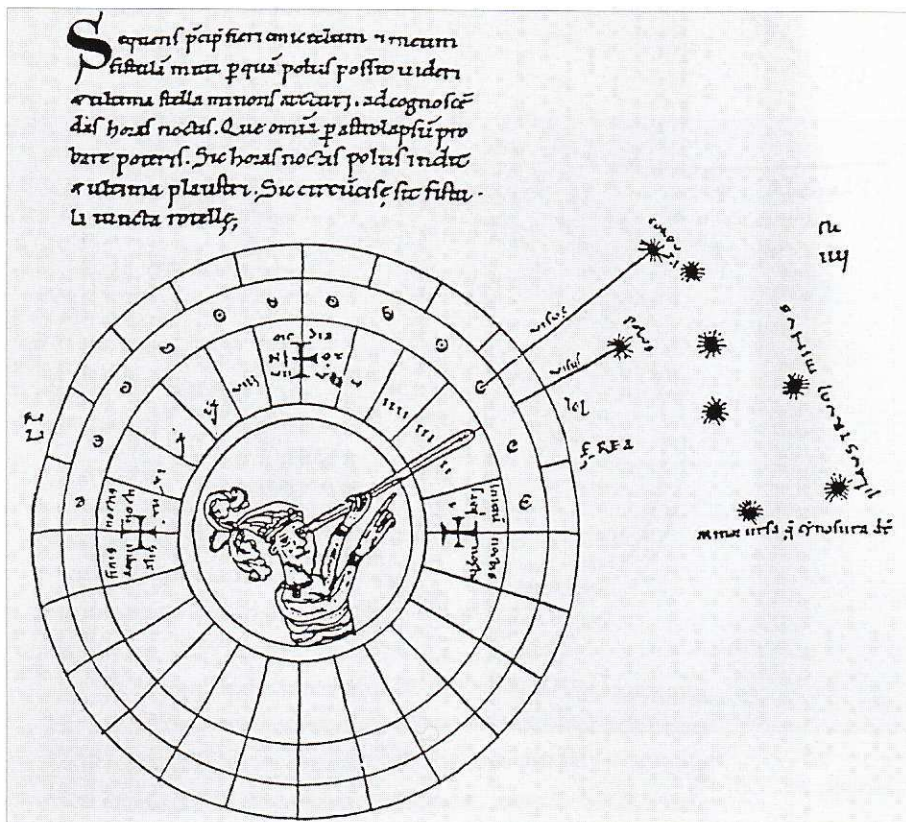


Fig.4 Chartres, MS 214 (olim 173), fol. ? Destroyed in 1944.

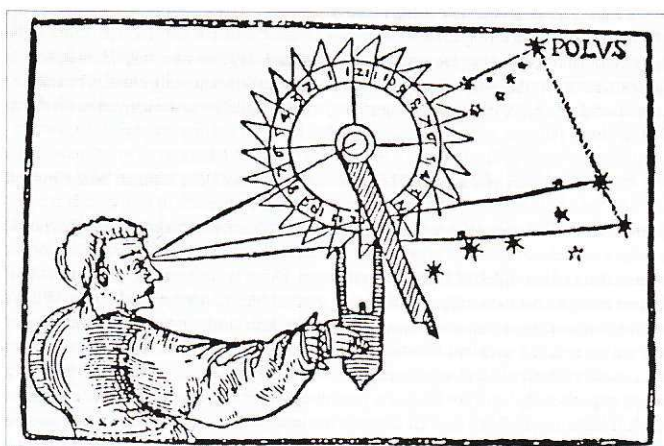


Fig.5 Peter Apian, Cosmographicus liber (1533).

of some conspicuous circumpolar stars about 1000 B.C. He postulated that prismatic jade tubes and disks were used in combination and formed an astronomical instrument for knowing the time by night. The agreement of star positions is not as closely as Michel stated, however. Moreover, he was not a sinologist and textual evidence stands against his speculations. Michel's claims, at first glance compelling and fascinating, have recently been refuted entirely by modern sinological research.²

In the first half of the 9th century a forerunner of the nocturnal was con-

structed in Italy. Pacificus (d. 844), head of the scriptorium of Verona Cathedral, claimed to be the inventor of the disk with sighting-tube.³ He described his *horologium nocturnum* in a poem⁴ and the sighting tube was depicted several times in manuscripts dating from the 11th and 12th centuries.⁵ In the 19th century these illustrations led to some extravagant claims for the presumed invention of the telescope in the Middle Ages (Figs1-4).

Because of the precession about 1000 A.D., the Pole Star stood at a distance of over 6° (corresponding to twelve-fold

diameter of the Moon) distance from the Celestial Pole. Around 800 A.D. the Celestial Pole was very near a star in the (of course much later formed) constellation Giraffe (32 Camelopardalis Hevelii). It is a star of only 4th to 5th magnitude, but it lies almost isolated and can be seen in clear nights. Only this star, which was also used as a guide star for navigation by the Vikings, could be observed by Pacificus.⁶ According to Pacificus a bright star near the Celestial Pole called by him *noctium horarum computatrix* was used for time-telling. Most certainly this was our Pole Star α Ursae minoris, a star of 2nd magnitude and in Pacificus' times over 7° off the Celestial Pole. Together with the water-clock and the astrolabe the *horologium nocturnum* served as a timekeeper to regulate monastic life.

In 997 Thietmar of Merseburg reported in his chronicle, that Gerbert of Aurillac had constructed a tube (*fistula*) and observed the guide star of the mariners (*stella nautarum duce*).⁷ At the end of the 13th century Raimundus Lullus described the nocturnal, which he called *Astrolabium nocturnum*.⁸

Contrary to the astrolabe the manuscript tradition of the nocturnal does not seem to be very broad. It may be noted, that no Islamic examples of this type of instrument are known. Eighteen manuscripts dating from the 15th and 16th centuries treating either the construction or use of the nocturnal (or both) are known.⁹ Most of these texts are part of a collection of astronomical texts or tables. Perhaps a comprehensive examination of dialling manuscripts would bring more texts to light.

No list of extant nocturnals is available. By a rather superficial census based only on examination of printed catalogues and inventory lists of German, British and American collections 74 nocturnals up to the year 1800 could be located. From these 25 are made of wood (mostly boxwood). A great number of nocturnals can be found in combination with other instruments (astronomical compendiums and quadrants).

The first printed description (Fig. 5) seems to be included in Peter Apian's *Cosmographicus liber* (Landshut 1524).¹⁰ The book was enlarged by Gemma Frisius in 1529 and until 1609 it went through 30 editions and several translations¹¹ thus, certainly contributing to the spread of the instrument. The nocturnal appeared also in Apian's *Instrumentbuch* of 1533 where it was combined with a universal dial of the Regiomontanus type and called *Horometer*: [note]¹² woodcuts to be glued on wood or pasteboard were

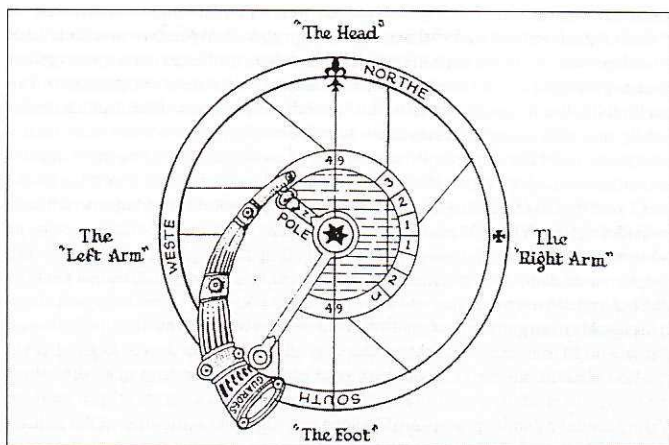


Fig. 6 Martin Cortes, *The Arte of Navigation* (1561).

added as a substitute for expensive instruments made of metal. Only very few of these instruments have survived. They were literally worn out in daily use.

Johannes Dryander (Eichmann, 1500-1560)¹³ was the author of a second description of the nocturnal after Apian. Dryander studied in Erfurt and in 1535 he was appointed as Professor of Mathematics and Medicine at Marburg University. Besides his medical activities - as one of the first Dryander introduced courses in practical anatomy - he occupied himself with the construction of astronomical instruments and wrote several small treatises. Some of these booklets are very scarce today. In 1533 Dryander was conferred a doctorate in Mainz and he entered the service of Johann III, Archbishop of Trier. At this time Dryander got into close contact with the mathematician and printer Johann Köbel of Oppenheim (c. 1462-1533), who edited several publications by Johannes Stoeffler and Sebastian Münster.¹⁴ Dryander published Köbel's unfinished treatise on the nocturnal (*Das Nocturnal Oder Die nachtvhr*, Frankfurt/M. 1535) of which only two complete copies are extant.¹⁵ There is also a description of the construction and use of the nocturnal by Sebastian Münster (1488/89-1552) included in his book on dialling (Fig. 7). The earliest English text seems to be contained in Thomas Fale's *Horologiographia* (London 1593).¹⁶

Closely connected with the history of the nocturnal are certain diagrams in nautical texts, which served as mnemonic devices for the correction of the measured altitude of the Pole Star. The position of α and β Ursae minoris respectively α and β Ursae maioris, often called 'Guards', indicated the correction to reduce the observed Pole Star altitude to obtain latitude. A man with stretched-out arms standing in the Pole was used. If the Guards were found over his head,

the Pole Star stood $3^{\circ}5'$ under the Celestial Pole and *vice versa*. The two arms marked the side deviations and also intermediate positions were recorded in mnemonic verses.¹⁷ This diagram was transferred into an instrument by Martín Cortés in his *Arte de Navegar* (written 1545, published in 1551). It consisted of a plate with the four points of the compass, a scale which indicated the amount of correction for the measured altitude of the Pole Star, and a pointer in the form of a trumpet on which seven stars of the Lesser Bear were marked (Fig. 6). The use of this instrument was similar to the nocturnal: The Pole Star was observed through the sighting hole, then the pointer was aligned with the Guards and the correction could be read off directly. But the use of the instrument lead to erroneous results, because the pole distance Cortés used was false. He adopted a value of $4^{\circ}9'$ from the Nuremberg astronomer Johann Werner (1541), despite in the text he mentioned another one used by mariners ($3^{\circ}30'$), which was much better.¹⁸ The reason for this is worth quoting:

The altitude is taken of the North starre: which is a starre in the extremitie or ende of the tayle of the lesse Beare, being a constellation commonly called the Home. [...] This starre hath declination 85. degrees, and 51. minutes. And the complement to 90.

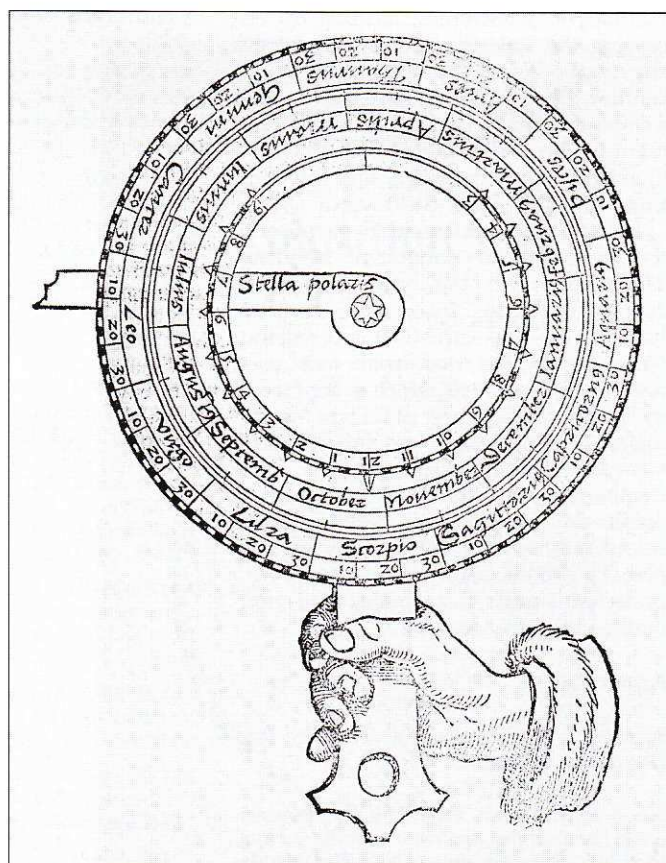


Fig. 7 Sebastian Münster, *Fürmalung vnd künstlich beschreibung der Horologien ...* (Basel, 1537), p. 147.

(whiche are 4. degrees and 9. minutes) is his distance from the Pole. And although the Maryners holde opinion that it is not distant more then three degrees & a halfe: yet to my iudgement, more credit ought to be geuen to the Astronomers then to Maryners: Forasmuch as the Astronomers do knowe the places of the starres [...] more perfectly & precisely then do the Maryners. For they account not onely by degrees, but also by minutes and secundes.¹⁹

In the 17th century also Tables for the correction of the altitude of the Pole Star were compiled.²⁰

The nocturnal was used by seafarers for altitude correction and time-reckoning: *Nocturlabe est vn instrument par lequel - ... toute heure de la nuit on peut trouuer combien l'Etoile du Nord est plus haute ou plus basse que le Pole. On peut aussi servir pour savoir quelle heure il est*, as Georges Fournier put it in his *Hydrographie*, printed in Paris 1643.²¹ It was also a useful instrument for computing the time of High Water at any place for which the 'establishment' (i.e. the interval of time which elapses between the time of meridian passage of Full or New Moon and the time of the following High Water) was known.²² Scarcely one nautical textbook omits a description of the nocturnal²³, but contrary to the obvious widespread use of the instrument the

method of determining latitude by observing the Pole Star disappeared from the textbooks at the end of the 17th century. Despite it was more convenient to observe the Pole Star at night than taking a direct sighting of the sun with a Quadrant or Cross-Staff, the Development of the Back-Staff and Davis' Quadrant made it possible to observe the sun without danger of blinding. There was also the need to rework correction tables from time to time because of the effect of precession. Nevertheless the nocturnal was used until the 18th century, which is confirmed by the great number of extant instruments. Certainly there was a discrepancy between learned authors of nautical treatises, who often were inclined to emphasize precise and mathematically sound methods for determining geographical positions, and common users, who preferred simple instruments and non-mathematical procedures.

Appendix

Manuscripts Treating the History and Use of the Nocturnal

(Zi. = Ernst Zinner, *Verzeichnis der astronomischen Handschriften des deutschen Kulturgebietes*, München 1925 (Additions: 'Aus alten Handschriften', *Bericht der Naturforschenden Gesellschaft Bamberg*, 38 (1962), pp. 8-57 by the same author).

- 1.) Zi. 7464
Wolfenbüttel, HAB: Cod. Guelf. 81.26 Aug. 2^o, fol. 144^v (Use of the Nocturnal, Latin, 1461)
- 2.) Zi. 3730
Göttingen, UB: 2^o Phil. 42m, fol. 55^v.00 (Johann v. Gmunden [?], Construction of the Nocturnal, in a collection of astronomical texts, 15th cent.)
- 3.) Zi. 7472
Würzburg, UB: M. ch. q. 132, fol. 153^v-154^v, 155^v (Construction of the Nocturnal, Latin, in a collection of astronomical texts, late 15th cent.)
- 4.) Zi. 7467
Leipzig, UB: Cod. 1469, fol. 201^r-207^r (Construction of the Nocturnal, Latin, in a collection of astronomical texts, 14/15th cent.)
- 5.) Zi. 7465
Munich, Bayer. StB: Clm 9092, fol. 262^v-264^v (Construction of the Nocturnal, Latin, in a collection of astronomical texts, 1679)
- 6.) Zi. 7469
Munich, Bayer. StB: Clm 24105, fol. 65^v-67^r (Use of the Nocturnal, German, in a collection of astronomical texts, 15/16th cent.)
- 7.) Zi. 7471
Munich, Bayer. StB: Clm 214, fol. 167^r-185^v (Construction of the Nocturnal, Latin, in a collection of astronomical tables, 15th cent.)
- 8.) Zi. 674
Munich, UB: 4o Cod. ms. 876, fol. 62^r-81^v (*De fundamento horario gnomonica*; therein: Construction of the Nocturnal, Latin, 1599 (Records of mathematical lectures given by the Jesuit Johannes Appenzeller in Ingolstadt))
- 9.) Zi. 7466
Vienna, ÖNB: Cod. Vin. 5228, fol. 98^v-105^r

(Construction of the Nocturnal, Latin, 16th cent.)

- 10.) Zi. 7470
Vienna, ÖNB: Cod. Vin. 5176, fol. 47^v-53^r (Construction of the Nocturnal, Latin, 16th cent.)
 - 11.) Zi. 9907
Vienna, ÖNB: Cod. Vin. 5292, fol. 312^r-313^v (Johann Stabius, Construction of the Nocturnal, Latin, 16th cent.)
 - 12.) Zi. 7468
Bern, Burgerbibl.: Cod. 157, fol. 27^v-28^v (Construction and Use of the Nocturnal, Latin, 15th cent.)
 - 13.) Zi. 11716
Zürich, Zentralbibl.: Ms. C 107, fol. 107^{r/v} (Wilhelm Hofer, (Carthusian monk and pupil of Georg Peuerbach), Construction and Use of the Nocturnal, Latin, Gaming (Lower Austria) 1472/79; see L. C. Mohlberg, *Mittelalterliche Handschriften* (= Katalog der Zentralbibliothek Zürich, vol. I), Zürich 1951, p. 55f., 361).
 - 14.) Zi. 2843b, c Würzburg, UB: M.ch.f.201
Construction of Sundials, Quadrants, Astrolabes and Nocturnals, 1599/1605 HG, Regensburg
 - 15.) Zi. 1345i Augsburg, Diözesanarchiv: 81/I Veit Bild (b. 1481), Augsburg, On the Cross-Staff and the Nocturnal
 - 16.) Zi. 7468a
Ottobeuren, Klosterbibl.: Ms. II, 319, fol. 122-123 (Construction of the Nocturnal, 15th cent.)
 - 17.) Zi. 9049a
Meiningen, Landesbibl.: Pd 32.44, fol. 92^r-98^v, 104^r-105^v (Construction of the Nocturnal, 15th cent.)
- ## Notes and References
1. 'Les jades astronomiques chinois, une hypothèse sur leur usage', *Bulletin des Musées Royaux d'Art et d'Histoire Bruxelles*, No. 1-3, (1947), pp. 31-38; 'Les jades astronomiques chinois', *Communications de l'Académie de Marine*, 4 (1947/49), pp. 111-133; 'Chinese Astronomical Jades', *Popular Astronomy: A Review of Astronomy and Allied Sciences*, 58 (1950), pp. 222-230; 'Sur les jades astronomiques chinois', *Mélanges Chinois et Bouddhiques*, 9 (1951), pp. 153-160; 'Le plus ancien instrument d'astronomie: Le Pi', *Ciel et Terre: Bulletin de la Société Belge d'Histoire, de Météorologie et de Physique du Globe*, 75 (1959), pp. 203-213.
 2. Christopher Cullen and Anne S. L. Farrer, 'On the term Hsüan Chi and the flanged trilobate jade disks', *Bulletin of the School of Oriental and African Studies*, 46 (1983), pp. 52-76. I am indebted to Marvin Bolt (Adler Planetarium, Chicago) for drawing my attention to this important article.
 3. See the recent contributions by Joachim Wiesenbach, 'Pacificus von Verona als Erfinder einer Sternenuhr', in P. L. Butzer und D. Lohrmann (eds.), *Science in Western and Eastern Civilisation in Carolingian Times* (Basel, 1991), pp. 229-250 and 'Der Mönch mit dem Sehrohr: Die Bedeutung der Miniatur Codex Sangallensis 18', p. 45' in *Schweizerische Zeitschrift für Geschichte*, 44 (1994), pp. 367-388; also Henri Michel, 'Les tubes optiques avant le telescope', *Ciel et Terre: Bulletin de la Société Belge d'Histoire, de Météorologie et de Physique du Globe*, 70 (1954), pp. 175-184.
 4. Text und German translation: Wiesenbach, 'Pacificus' (note 3), p. 237.
 5. Chartres Ms 214 (olim 173; destroyed in 1944); Venedig, Bibl. Naz. Marciana, Ms. lat. VIII.22, fol. 1^r; Rom, Bibl. Apostolica Vaticana, Vat. lat. 644, fol. 76^r; Stiftsbibl. St. Gallen, Cod. Sang. 18, p. 45; Avranches, Bibl. Municipale, Ms. 235, fol. 32^r.
 6. Wiesenbach, 'Pacificus' (note 3), pp. 244sq.
 7. Wiesenbach, 'Pacificus' (note 3), p. 230.
 8. Raimundus Lullus, Opera, Mainz 1721/42, vol. I (*Liber principiorum medicinae*), pp. 46f. (written on the island of Majorca, 1273/75).
 9. See the list in the appendix.
 10. I was not able to locate a copy of a text by Martin Waldseemüller printed in 1507 (*Declaratio horarii novi varios habentis effectus*) mentioned by Joseph Drecker in his *Zeitmessung und Sterndeutung in geschichtlicher Darstellung* (Berlin, 1925), p. 145.
 11. Fernand van Ortoy, *Bio-Bibliographie de Gemma Frisius fondateur de l'école Belge de géographie de son fils Corneille et de ses neveux les Arsenius* (Brussels, 1920, new ed. Amsterdam, 1966), p. 163-189.
 12. Peter Apian, *Instrument Buch/ durch Petrum Apianum erst von neu beschriben* (Ingolstadt, 1533), fol. G IIIr-H IIIv; see also Albert Schück, 'Das Horometer, ein älteres Instrument der mathematischen Geographie', *Mitteilungen der Geographischen Gesellschaft in München*, 1 (1904/06), pp. 269-283, who describes a Quadrant with Regiomontanus's Universal Dial and Nocturnal by Caspar III Buschmann dated 1611 (Berlin, Kunstgewerbemuseum: Inv.-Nr. P 29).
 13. Petra Schachtner, 'Johannes Dryander und die Aufwertung der angewandten Mathematik zur Universalwissenschaft', in: Barbara Bauer (ed.), *Melanchthon und die Marburger Professoren (1527-1627)* (= Schriften der Universitätsbibliothek Marburg, 89), Marburg, 1999, vol. II, pp. 790-821; Wolf-Dieter Müller-Jahncke, 'Johannes Dryander und Heinrich Cornelius Agrippa von Nettesheim in ihrem Briefwechsel', *Hessische Heimat*, 4 (1975), pp. 91-98.
 14. Josef Benzing, *Jakob Köbel zu Oppenheim 1494-1533: Bibliographie seiner Drucke und Schriften* (Wiesbaden, 1962), Nos. 27, 58, 59, 89, 91, 93.
 15. Located in the Stiftsbibliothek of Aschaffenburg (E-470/Bb.3) and in Göttingen University Library (8^o Astron. II, 1733; bound together with Johannes Schoener, *Tabulae astronomicae*, Nürnberg, 1536). A copy of a Latin version (*Nocturnalis Instrumenti, per quod horae nocturnae addiscuntur, Compositio & Vsus, ex uarijs autoribus, in gratiam Astronomiae studiosorum, congesti* (Marburg, 1543); see Arrey von Dommer, *Die ältesten Drucke aus Marburg in Hessen 1527-1566* (Marburg, 1892), p. 179) seems not to be preserved. Köbel's text including the woodcuts was reprinted several times since 1578 in the *Astronomia Teutsch* (fol. 41-47), a popular astronomical-astrological compendium.
 16. Thomas Fale, *Horologographia. The Art of Dialling: teaching an easie and perfect way to make all kinds of Dials vpon any plaine plat howsoever placed [...]* (London, 1593). Fale studied 1578/97 in Cambridge and in 1604 was given permission to set up as a doctor.
 17. See for example the *Regimento do Norte* (probably composed in the 15th century; München, Bayer. StB: 4^o Inc., 1551). On the nautical 'Regiments' see Joaquim Bensaude,

L'astronomie nautique au Portugal à l'époque des grandes découvertes (Bern, 1912/17, repr. Amsterdam 1967), pp. 136-145, 223f.; Hermann Wagner, 'Die Entwicklung der wissenschaftlichen Nautik im Beginn des Zeitalters der Entdeckungen nach neuern Anschauungen', *Annalen der Hydrographie und Maritimen Meteorologie*, 46 (1918), pp. 215-220.

18. David W. Waters, *The Art of Navigation in England in Elizabethan and Early Stuart Times*, 2nd ed. (Greenwich, 1978), part I, pp. 45f.

19. Martín Cortés, *The Arte of Nauigation, Conteynyng a compendious description of the Sphere, with the making of certain Instrumentes and Rules for Nauigations: and exemplified by manye Demonstrations*, Transl. Richard Eden

(London, 1561), fol. 64^r.

20. Hans-Christian Freiesleben, 'Der Polarstern in der Navigation', *Sudhoffs Archiv für Geschichte der Medizin und der Naturwissenschaften*, 39 (1955), pp. 157-161; Id., 'Early Pole Star Tables', *Journal of the Institute of Navigation*, 8 (1955), pp. 285-288; J. B. Hewson, *A History of the Practice of Navigation*, 2nd ed. (Glasgow, 1983), p. 118.

21. Quoted from Francis Maddison, 'Medieval Scientific Instruments and the Development of Navigational Instruments in the XVth and XVIth Centuries', *Revista da Universidade de Coimbra*, 24 (1971), p. 145.

22. Charles H. Cotter, *A History of Nautical*

Astronomy (London/Sydney/Toronto, 1968), p. 40.

23. E.g. Edward Wright, *Certaine Errors in Navigation* (London, 1610), part I, p. 210sq.; John Seller, *Practical Navigation; or, An Introduction the whole Art [...]* 4th ed. (London 1680), pp. 173-178 (first ed.: 1669).

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Anecdotal Instruments: An Unusual Electrical Tester

Paolo Brenni

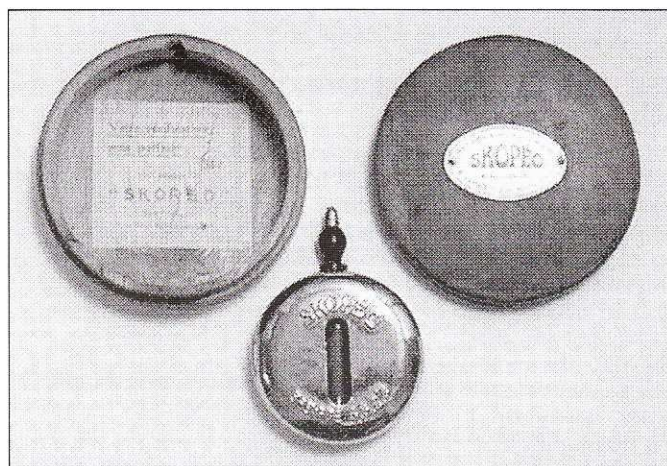


Fig.1 The SKOPÉO sparking plug tester and box.



Fig.2 The opened case revealing the small neon tube.

In spite of the fact that Parisian flea markets are flooded with fake late 19th century and early 20th century English apparatus manufactured in Asia, it is still possible to discover some unusual (and cheap) instruments if you keep your eyes wide open. Very recently in a small temporary market I noticed a round wooden box with a metallic label bearing the following inscription:

Rayon X de l'Allumage
SKOPÉO
Marque déposée Breveté S.G.D.G
et tous Pays.

This contained a pocket watch shaped instrument (diameter about 6 cm) made of nickel-plated metal (Figs 1 and 2) with a short protruding isolated electrode. A

vertical window of the case shows a small glass phial. After having opened the case I saw a small vacuum tube connected to the electrode and held by a metallic elastic spring. In the wooden box a second label was inscribed: 'Votre vérification sera parfaite SI? Vous employez «SKOPÉO» dans l'obscurité et votre moteur au ralenti'. These few indications gave me a hint about the use the device which, in spite of the advertising, has nothing to do with X-rays. In fact, the pocket instrument was used to check the good functioning of spark plugs in an era when car engine were not compact black boxes and sparking plugs (less reliable than today) were easy reachable. Instead of using the 'classical' screwdriver for making a short spark gap between the external electrode of the plug and the

mass of the engine, one could simply touch the conductor (or the wire) of the plug with the electrode of the instrument. Every spark produced by the plug in the cylinder corresponded to a flash of the evacuated tube. I bought the little instrument for about £12. As soon as I arrived at home with the 'SKOPÉO' I tested it with the spark of a piezoelectric lighter. To my great satisfaction, the tube produced a red flash typical of neon lamps (*Glimmlicht*). Neon lamps were widely used for signals and advertising. The spark plug testing apparatus had probably been manufactured in the 1930s but even today you can find screwdriver-shaped plug tester that works on the same principle.